

TITLE PAGE

- **Problem Statement ID** – Question 5
- **Problem Statement Title-** Grade Change Intelligence for Paper Manufacturing
- **Theme-** Artificial Intelligence & Machine Learning
- **PS Category-** Software
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- **Student ID** – 23BCE0444

PaperAI – AI-Powered Grade Change Intelligence

Proposed Solution: PaperAI is an AI-powered decision support system that predicts **OFF-SPEC paper production during grade transitions** before quality limits are exceeded. The platform analyzes real-time machine parameters using a **Random Forest model**, compares them with **historical successful production runs**, recommends corrective machine setpoints, and explains every recommendation using **Google Gemini**.

How it addresses the problem

- Predicts OFF-SPEC production before paper quality deteriorates.
- Continuously monitors Basis Weight deviation ($\pm 2.5\%$ tolerance).
- Recommends corrective machine settings using historical successful runs.
- Reduces stabilization time and production losses.
- Provides explainable AI recommendations for operators.
- Records operator feedback for continuous improvement.

Innovation & Uniqueness

- Machine Learning-based OFF-SPEC prediction.
- Historical trajectory comparison with successful production runs.
- Explainable AI using Google Gemini.
- Confidence score for every recommendation.
- Interactive analytics dashboard with Digital Twin.
- One-click PDF & CSV report generation.
- Human-in-the-loop operator feedback system.

TECHNICAL APPROACH

Technologies Used

Frontend: Streamlit

Machine Learning: Random Forest (Scikit-learn)

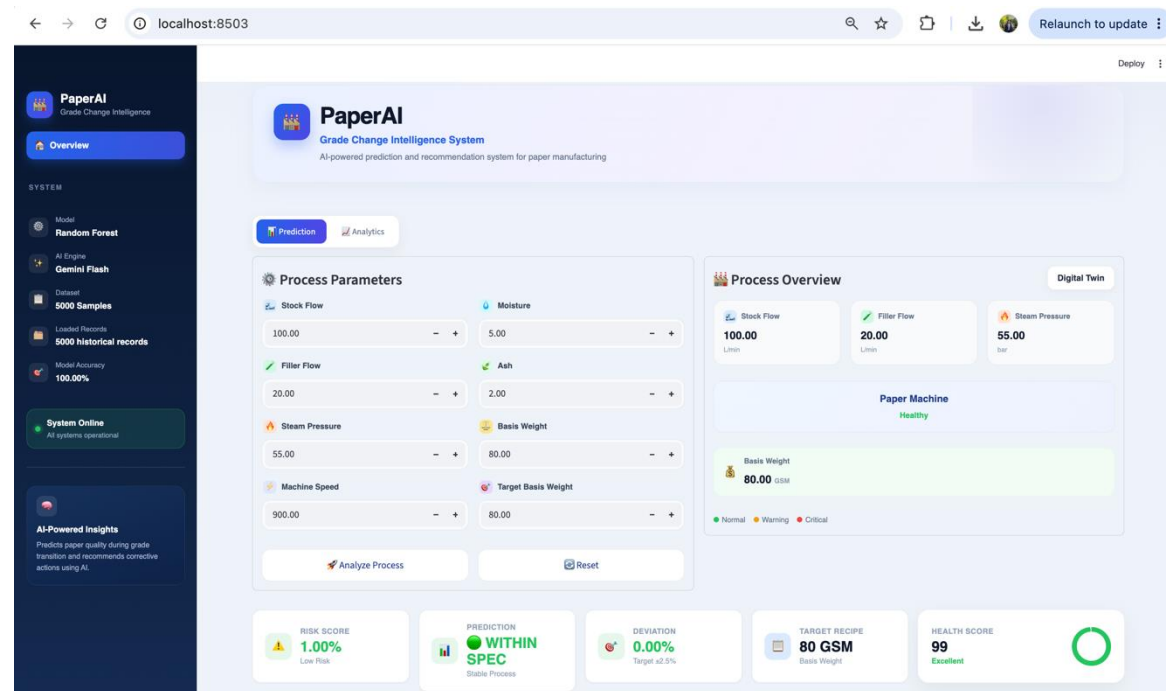
Explainable AI: Google Gemini Flash

Data Processing: Pandas, NumPy

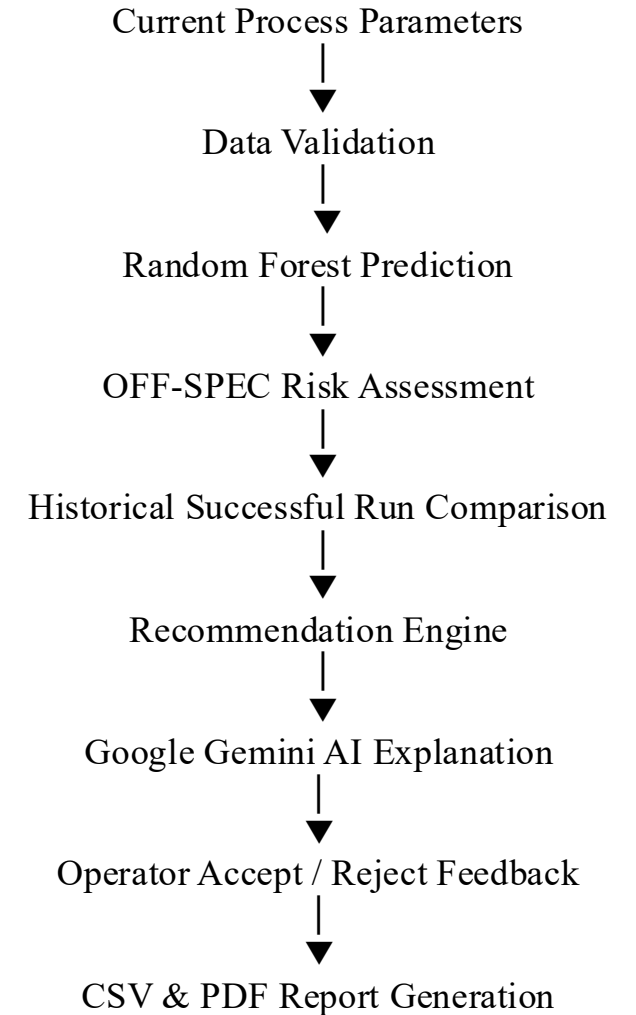
Visualization: Plotly

Model Storage: Joblib

Report Generation: ReportLab (PDF), CSV



Methodology and process for implementation



FEASIBILITY AND VIABILITY

Feasibility of the Solution

- Uses historical production data already available in QCS/DCS systems.
- Built with Python, Streamlit, Random Forest, and Google Gemini, making it easy to deploy.
- Requires no additional hardware and can integrate with existing manufacturing workflows.
- Recommendations are generated in real time, enabling proactive operator intervention.

Potential Challenges & Risks

- Model accuracy depends on the quality and quantity of historical process data.
- Sensor failures or missing data may affect prediction reliability.
- Different paper grades may require periodic model retraining.
- Initial operator adoption may require validation and trust building.

Strategies to Overcome Challenges

- Continuously retrain the model using new production data and operator feedback.
- Validate sensor inputs and handle missing values before prediction.
- Use Explainable AI (Gemini) to provide transparent reasoning for recommendations.
- Enable operator Accept/Reject feedback to improve recommendations over time.

ARTIFACTS

Relevant Artifacts

Source Code (Python + Streamlit)

Trained Random Forest Model (random_forest.pkl)

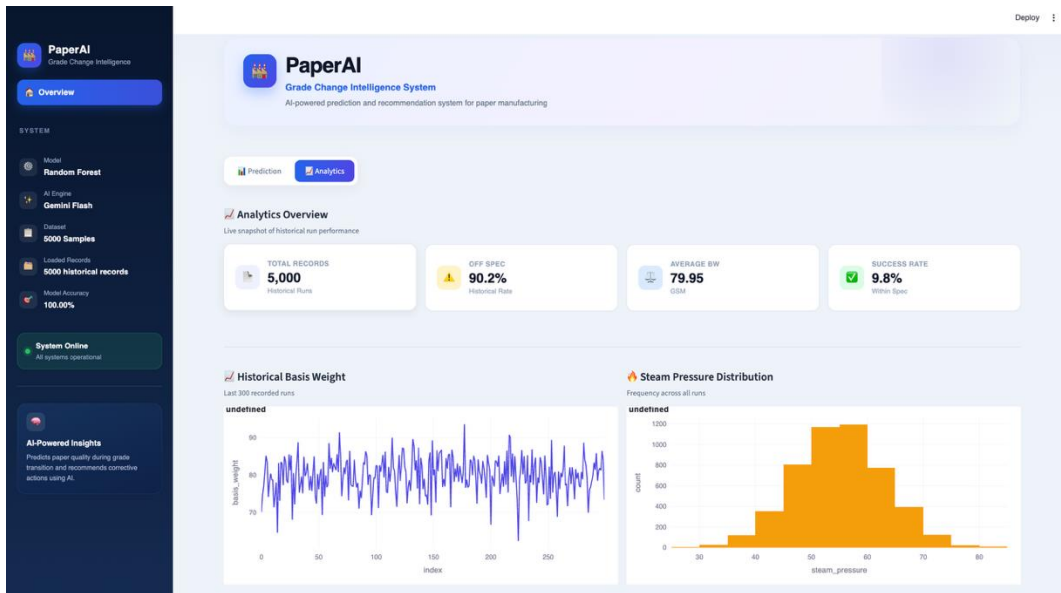
Historical Production Dataset (historical_data.csv)

Dashboard Screenshots

Project Documentation (README.md)

Requirements File (requirements.txt)

Feedback Log (feedback.csv)



```
EXPLORER
PAPERAI
  data
    historical_data.csv
  models
    accuracy.pkl
    confusion_matrix.png
    feature_importance.png
    random_forest.pkl
  tests
    list_models.py
    test_api.py
    test_gemini.py
    test_new.py
  venv
  .env
  .gitignore
  app.py
  feedback.csv
  feedback.py
  gemini_helper.py
  generate_data.py
  README.md
  recommendation.py
  requirements.txt
  train.py

app.py
1 import streamlit as st
2 import pandas as pd
3 import joblib
4 import plotly.express as px
5 import plotly.graph_objects as go
6 import os
7 from datetime import datetime
8 from reportlab.platypus import SimpleDocTemplate, Paragraph
9 from reportlab.lib.styles import getSampleStyleSheet
10 from io import BytesIO
11
12 from gemini_helper import explain
13 from feedback import save_feedback
14 from recommendation import get_recommendations
15
16 st.set_page_config(
17     page_title="Paper Grade Intelligence",
18     page_icon="📄",
19     layout="wide"
20 )
21
22 if "analysis_done" not in st.session_state:
23     st.session_state.analysis_done = False
24
25 model = joblib.load("models/random_forest.pkl")
26 history = pd.read_csv("data/historical_data.csv")
27
28 st.html("""
29 <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap"
30 <style>
31 :root{
32     --ink:#0F172A;
```

RESEARCH AND REFERENCES

Research Sources

- Honeywell Hackathon Problem Statement – *Grade Change Intelligence in Paper Making Process*
- Honeywell Quality Control System (QCS) documentation and process description
- Scikit-learn Documentation (Random Forest Classifier)
<https://scikit-learn.org/stable/>
- Streamlit Documentation
<https://docs.streamlit.io/>
- Plotly Documentation
<https://plotly.com/python/>
- Google Gemini API Documentation
<https://ai.google.dev/>
- Pandas Documentation
<https://pandas.pydata.org/docs/>
- NumPy Documentation
<https://numpy.org/doc/>